

Weather Market Trading Anomaly Explorer

Research methods and system report

Polymarket and Kalshi weather prediction markets

Prepared for academic review

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Purpose

This report explains what the application does, how it calculates each score, and how to read the results. The system finds unusual public trading activity that may deserve human review. It does not decide that fraud or illegal trading occurred.

Main point

The application turns a large number of public trades into a shorter list of cases to review. Each case shows the trades, calculations, signal scores, and missing information. Unusual activity is not proof of wrongdoing. Kalshi also does not provide public trader identities.

Platform	Unit of analysis	Identity evidence	Appropriate conclusion
Polymarket	Wallet market episode	Pseudonymous public wallet	Which wallet episodes deserve review
Kalshi	Trade and grouped market activity event	No public trader identity	Which market activity periods deserve review

Purpose of the project

The project asks a simple question: which trades in open weather prediction markets look unusual enough for a researcher to review? It collects public market data, groups related trades, calculates five signals, and shows the strongest cases in a dashboard.

A higher score means that several parts of the trading pattern are unusual compared with similar activity. The score is not the probability of fraud. A researcher must still check market rules, public news or weather updates, data quality, and other possible explanations.

System workflow

Stage	What the system does	Research value
1	Finds open weather markets and removes nonweather markets	Keeps the project focused
2	Downloads public markets, prices, and trades	Creates a record of the source data
3	Standardizes times, outcomes, prices, sizes, and dollar values	Makes trades easier to compare
4	Groups related activity into Polymarket episodes and Kalshi events	Stops each small trade from becoming its own case
5	Calculates five signals and checks data reliability	Shows why an activity pattern looks unusual
6	Uses strict rules before showing a flag	Reduces weak or misleading flags
7	Shows graphs, calculations, limits, and source links	Helps a researcher review the evidence

Important terminology

Term	Meaning in this project
Market	A question people can trade on; its price shows the market's current view of the outcome
Wallet	A public pseudonymous blockchain address on Polymarket; it is not a verified personal identity
Trade notional	The dollar value of an executed trade, calculated as price multiplied by contracts or shares
Episode	One wallet's activity in one Polymarket market, separated from later activity by a 24 hour inactivity gap
Kalshi activity event	Related trades in one contract grouped until a 30 minute inactivity gap
Signal score	A value from 0 to 1; a higher value means that part of the activity is more unusual

Term	Meaning in this project
Reliability weight	A value that lowers or removes a signal when the supporting data are weak or missing

$$\text{Notional} = \text{Price} \times \text{Contracts}$$

Equation 1 Trade notional

Polymarket wallet episode model

Polymarket includes public wallet addresses with its trade data. The application can group one wallet's trades in one market and compare that episode with other traders and with the wallet's earlier activity. The five signals are called S1 through S5.

Signal	Question	Calculation and interpretation
S1 Size	Is this episode unusually large	Compares the dollar amount with other episodes in the market and with the wallet's earlier episodes. The larger comparison score is used.
S2 Timing	Did the trade happen at an unusual time	Compares the trade time with a verified public weather update when one is available. A weaker close-time estimate is used when verified timing is missing.
S3 Past results	Has the wallet done unusually well before	Uses earlier resolved results. The score stays close to neutral when the wallet has little history.
S4 Market share	Did the wallet make up a large share of activity	Combines the market's YES versus NO imbalance with the wallet's share of observed trading. Very small markets are excluded.
S5 Behavior	Does the overall pattern differ from similar episodes	Uses an Isolation Forest model to compare size, market share, direction, and earlier activity with similar weather episodes.

$$S_1 = \max(P_{\text{market}}, P_{\text{wallet}}), S_3 = 0.5 + 0.5 \tanh(2 \text{ROI} \lambda), S_4 = (|I| + P_{\text{share}}) / 2$$

Equation 2 Selected Polymarket signal formulas

Reliability controls

- S2 receives full weight only when a trusted public timestamp supports the timing comparison.
- S3 stays closer to neutral when only a few earlier wallet results are available.
- S4 is not used when the market has fewer than 20 trades or less than 1000 dollars in observed trading.
- A missing signal is left out instead of being scored as zero. Zero would incorrectly suggest normal activity.

Composite anomaly score

The system does not simply average the five signals. It changes each available signal into log odds, gives weaker data less weight, and combines the results. The logistic function then changes the result back to a score between 0 and 1. Several strong signals together raise the score, while weaker or opposite signals lower it.

$$\text{Composite} = \sigma(\sum_{i=1}^5 Q_i \log(S_i / (1 - S_i))), \sigma(x) = 1 / (1 + e^{-x})$$

Equation 3 Reliability weighted evidence combination

In the formula, S_i is one signal score, Q_i is the reliability of that signal, and sigma changes the result back to the 0 to 1 scale. This is an anomaly score, not a probability of fraud.

Polymarket review thresholds

Requirement	Implemented rule
Composite score	At least 0.75
Valid evidence	At least four available modules
Strong evidence	At least three active signals at or above 0.75 in the current scoring code
Minimum activity	At least two trades and 100 dollars notional, or one episode of at least 5000 dollars
Priority band	Score at least 0.92, all five modules available, and high confidence timing evidence

Kalshi market activity model

Kalshi's public data does not include a trader or wallet identifier. The application therefore cannot identify a person or study one account's history. Instead, it looks only at visible market activity. Kalshi flags describe unusual trades or market events, not suspicious users.

Signal	Public evidence used	Main limitation
K1 Trade size	Compares the trade with similar trades and checks whether it is at least meaningful in dollars	A trade can be large compared with peers but still small in dollars
K2 Timing	Looks for a burst of trades and activity close to market closing	It receives less weight without a verified public weather-update time
K3 Price movement	Looks for a price jump followed by a move back	A price move does not show the trader's reason
K4 Market pressure	Measures the trade as a share of observed activity in the contract	Public data may not show every part of market liquidity
K5 Difference from peers	Compares size, price, contract count, and time gaps with similar weather markets	An unusual pattern does not explain its cause

$$K_1 = (P_{\text{peer}} + \min(\text{Notional} / 250, 1)) / 2, K_4 = \min(20 \times \text{Trade Notional} / \text{Contract Notional}, 1)$$

Equation 4 Selected Kalshi signal formulas

Kalshi queue and event grouping

A Kalshi trade is shown for review only when its total score is at least 0.95, its size score is at least 0.85, at least three signals are 0.80 or higher, and the trade value is at least 250 dollars. Related trades are grouped into one event when they occur less than 30 minutes apart.

A high evidence grade requires at least 1000 dollars in grouped trading, four strong signals, and a meaningful price movement. A medium grade requires at least 250 dollars and three strong signals. These grades help order the review list. They do not prove misconduct.

Evidence unavailable for Kalshi

Unavailable evidence	What cannot be concluded
Trader identity	The dashboard cannot identify a person or account
Account history	It cannot calculate repeat success by the same user
Counterparty links	It cannot detect coordinated accounts
Order submissions and cancellations	It cannot infer intent from unexecuted orders
Account ownership on both sides	It cannot perform a direct wash trading test
Full order book behavior	It cannot establish spoofing or layering

How to read a flagged case

Dashboard element	What to ask
Market question	What exact outcome is being priced, and what are the settlement rules
Progression graph	Did traded value cluster in time, and did the executed price move or reverse
Trade table	Does price multiplied by size reproduce the displayed notional
Signal table	Which signals are high, and which reliability weights weaken them
Timing reference	Is the public event timestamp verified, or is it only a close time proxy
Missing evidence	Which conclusions are impossible with the available public data
Cross market context	Do similar contracts genuinely share location, unit, time period, and settlement rules

Worked interpretation example

Suppose a Kalshi event contains 2500 dollars across several trades. Four of the five signals are above 0.80, and the price rises from 0.30 to 0.75 before falling to 0.45. The application may give this event a high evidence grade. This means the burst was large, the price move was unusual, and the event differed from similar activity. It does not tell us who traded, why they traded, or whether the activity was manipulative.

Research safeguards and limitations

- The system uses public data and does not try to discover the real people behind Polymarket wallets.
- Open markets can change after data is collected, so each case should include a collection time.
- A market price shows traders' current views. It is not a direct measurement of the true probability.
- Isolation Forest finds rare patterns. It does not explain why the pattern is rare.
- Public weather-update times must be verified. A market's close time alone is not proof of an information advantage.
- The current thresholds are research rules. They have not been tested against a confirmed fraud dataset.
- A comparison across platforms is only useful when the markets have similar rules, units, locations, and time periods.

Validation and reproducibility

The project saves the original data, uses stable record IDs, and keeps data collection separate from scoring. It also records the feature and model versions. The automated test suite has 18 passing tests covering data clients, trade grouping, calculations, scoring, Kalshi events, and basic application behavior. These tests show that the software runs consistently, but they do not prove that every flag is correct.

Improvements from the earlier version

Earlier version	Improvement made
Included many market categories	Now focuses only on weather markets and removes nonweather Kalshi results
Mixed open and resolved markets	Main research screens now focus on open or live markets
Used demo or synthetic records	Dashboard now uses live public Polymarket and Kalshi data instead of demo data
Polymarket and Kalshi were presented together	Each platform now has its own section, method, formulas, and limits
Flags could appear with weak visible evidence	Stricter score, signal-count, trade-size, and data-quality rules now control the review list
Kalshi lacked wallet information	A separate market-activity model now scores only the public evidence Kalshi provides
A flag was difficult to understand	Each case now shows why it was flagged, the trade progression graph, calculations, reliability, and missing evidence
The interface was crowded and hard to read	The layout now uses clearer labels, score keys, readable graph text, separate platform views, and smaller status indicators

Possible next improvements

Two changes would make the project stronger. First, verified weather-update times from sources such as NOAA or the National Weather Service would improve the timing signal. Second, a set of cases reviewed by people would help measure false positives and choose better score thresholds.